Credit Card Financial Performance & Risk Analytics Analytics

This Project focuses on analyzing customer credit card usage data to evaluate financial performance, customer behavior, and credit risk exposure. Using Power Bi, key financial indicators such as utilization ratio, delinquency rate, transaction growth, and customer churn were measured to support data-driven decision-making in a banking environment.

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Challenges

Banks and financial institutions manage millions of credit card customers but face significant challenges in balancing profitability with risk management.

Identifying High-Risk Customers

Detecting customers who may default on payments before approving credit

Understanding Behavior

Analyzing customer spending patterns and credit utilization trends

Measuring Retention

Monitoring customer churn and evaluating credit limit appropriateness

Maintaining Profitability

Predicting loan eligibility while minimizing risk exposure

Without proper analytics, banks may approve loans to risky customers or lose valuable customers to competitors.

Project Objectives

01

Analyze Customer Behavior

Examine credit card transaction data to understand financial patterns

02

Calculate Key KPIs

Develop important metrics for financial decision making

03

Identify Risk Customers

Use credit utilization and delinquency data to flag high-risk accounts

04

Measure Trends

Track customer growth and transaction patterns over time

05

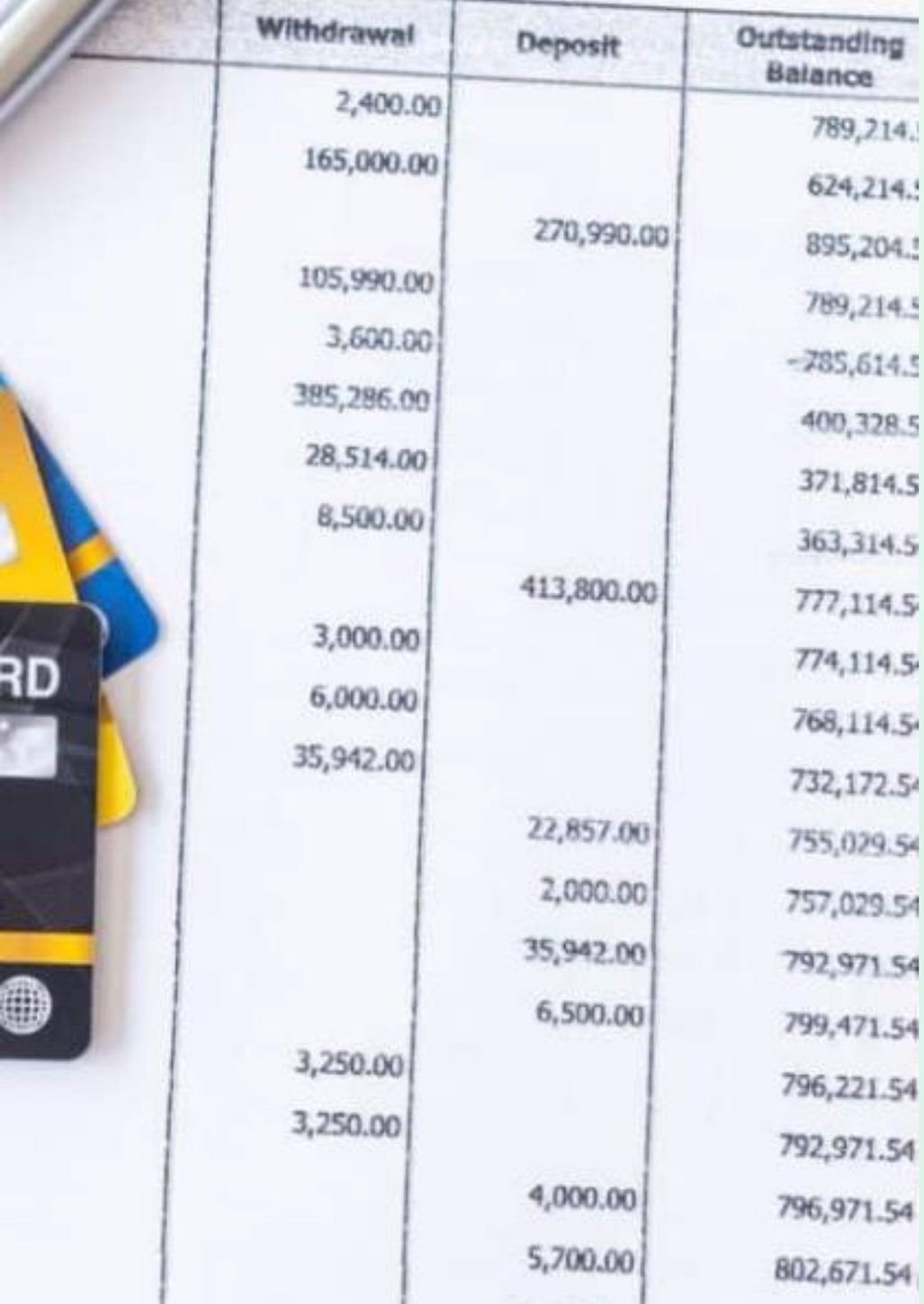
Evaluate Relationships

Analyze connections between income and credit limit

06

Support Decisions

Provide data-driven insights for business strategy



Dataset Overview

The dataset contains comprehensive customer financial and credit card information representing banking customer behavior.

Customer Table

- Client_Num
- Income
- Customer_Age
- Education
- Marital_Status
- Card_Category
- Personal_Loan

Credit Card Transaction Table Table

- Total_Trans_Amt
- Total_Trans_Ct
- Credit_Limit
- Avg_Utilization_Ratio
- Total_Revolving_Bal
- Delinquent_Acc
- Interest_Earned
- Cust_Satisfaction_Score

Tools & Techniques

Power BI Desktop

Primary visualization and dashboard platform for creating interactive financial reports

Power Query Editor

Data transformation and cleaning tool for preparing raw customer data

Data Cleaning

Removing null values, correcting data types, and handling missing information

DAX Calculations

Creating measures and financial ratios using Data Analysis Expressions language

Risk Scoring Model

Developing credit risk scores based on utilization and delinquency patterns

Key DAX Measures

Essential calculations used to analyze financial performance and credit risk.

1

Running Total of Credit Card Transactions

```
RUNNING_TOTAL_CREDIT_CARD =
CALCULATE(SUM(Credit_card[Total_Trans_Amt]),
Calendar[Date] <= MAX(Calendar[Date]))
```

2

4 Week Moving Average

```
4_WEEK_MOVING_AVERAGE =
VAR _last_date = LASTDATE(Calendar[Date].[Date])
RETURN AVERAGEX(DATEBETWEEN(Calendar[Date].[Date],
DATEADD(_last_date,-28,_last_date),
CALCULATE(sum(credit_card[Credit_Limit])))
```

3

Average Customer Satisfaction

```
Calculate(AVERAGE(Customers[Cust_Satisfaction_Score]),
GROUPBY(Credit_card,Credit_card[Card_Category]))
```

4

CAC Ratio

```
CAC_RATIO =
DIVIDE(SUM(Credit_card[Acquisition_Acq_Cost]),
SUM(Transaction[Total_Trans_Amt]))
```

Key DAX Measures

Essential calculations used to analyze financial performance and credit risk.

1

Top 5 By Transaction amount

```
top_5_by_transaction_amount =
TOPN(5,SUMMARIZE(Credit_card,Credit_card[Client_Num]
,”TotalAmount”,Sum(Credit_card[Total_trans_amt])),[T
otalAmount],DESC)
```

2

High Risk Flag (Credit_table)

```
High_risk_flag =
IF(Credit_card[Total_revolving_bal]>0.9*Credit_car
d[Credit_limit] &&
Credit_card[Avg_Utilization_Ratio] > 0.8,”High
Risk”,”Normal”
```

3

Average Utilization Ratio

```
Avg_Utilization_ratio =
IF(max(credit_card[Avg_Utilization_ratio])>0.8,”Abov
e_80%“,”Not”)
```

4

High Risk Flag (Measure)

```
High_risk_flag =
Calculate(COUNTROWS(Credit_card),Credit_card[High_ri
sk_flag] = “High Risk”)
```

Key DAX Measures

1

Yearly Average Utilization Ratio

```
YEARLY_AVG_UTILIZATION =
AVERAGE(Credit_card[Avg_Utilization_Ratio])
```

2

Percentage of Interest Earned

```
%_OF_INTEREST_EARNED =
DIVIDE(SUM(Credit_card[Interest_Earned]),
SUM(Credit_card[Total_Revolving_Bal]))
```

3

Delinquency Rate

```
DELINQUENCY_RATE = DIVIDE(COUNTROWS(Credit_card),
Credit_card[Delinquent_Acc] > 0),
COUNTROWS(Credit_card))
```

4

Average Credit Limit

```
Avg_credit_limit =
AVERAGE(Credit_card[Credit_Limit])
```

Key DAX Measures

Month-over-Month Growth

```
MOM_GROWTH =
VAR Pre_Mon_tran =
calculate(sum(Credit_card[Total_trans_amt]),DATEADD
(calendar[Date],-1,Month))
VAR cur_Mon_tran =
sum(credit_card[Total_trans_Amt])
RETURN DIVIDE([Cur_Mon_tran] - [Pre_Mon_tran],
[Previous Month Value])
```

Week-Over-Week Growth

```
WOW_GROWTH =
VAR Pre_week_tran =
calculate(sum(Credit_card[Total_trans_amt]),DATEADD
(calendar[Date],-7,Day))
VAR cur_week_tran =
sum(credit_card[Total_trans_Amt])
RETURN DIVIDE([Cur_week_tran] - [Pre_week_tran],
[Previous Month Value])
```

Key Dax Measures

Credit Risk Score

```
CREDIT_RISK_SCORE =
VAR Utilization_score =
AVERAGE(credit_card[Avg_Utilization_ratio]) *100
VAR Delinquency_score =
MAX(credit_card[Deliquent_acc])*20
VAR Revolving_balance_score =
DIVIDE(sum(credit_card[Total_revolving_bal]),100
0,0)
RETURN
Utilization_score + Delinquency_score +
Revolving_score
```

Customer Churn

```
Customer_churn =
VAR Churned_client =
Filter(SUMMARIZE(credit_card,credit_card[client_
num],”LastTransDate”,max(Calendar[Date])),
DATEDIFF([LastTransDate],max(Calendar[Date]),MON
TH) >= 6)
RETURN
      COUNTROWS(Churned_client)
```

Key DAX Measures

Income Credit Correlation

```
Income_credit_correlation =
VAR avg_income = AVERAGE(credit_card[Annual_Fees])
VAR Avg_credit_limit = AVERAGE(credit_card[Credit_limit])
VAR numerator =
Sum(credit_card,(credit_card[annual_fees]-avg_income) *
(credit_card[credit_limit]-Avg_credit_limit))
VAR denominator =
SQRT((SUMX(Credit_card,POWER(Credit_card[Annual_fees]-
avg_income,2)) *
SUMX(credit_card,POWER(credit_card[credit_limit]-
avg_credit_limit,2))))
RETURN DIVIDE(numerator,denominator,0)
```

Implementation Methodology

Data Cleaning (Power Query)

- Removed null values
- Corrected data types
- Handled missing credit limits and utilization ratios
- Removed duplicate customer records

Data Modeling

- Created relationships between Customer and Credit_card tables using Client_Num
- Built star schema model for efficient querying

DAX Calculations

- Created measures for KPIs
- Calculated financial ratios
- Developed risk scoring formula

GitHub.com LinkedIn.com

Video Based Project Performance Overview

Also Available on

LinkedIn

GitHub

Google Drive

Click on Link or given site

https://drive.google.com/drive/folders/1EnCLZeTZ2EwLG2owmfgcXdG4mvNyzo5u

Wscube_Tech

Key Findings

46M

Total Transactions

Amount processed across all customer accounts

6.06%

Delinquency Rate

Percentage of accounts with late payments

112

High Risk Customers

Identified through risk scoring model

66.63%

Interest Earned

Ratio of interest collected from total credit

Key Findings

1.67M

4_Week_moving_average

The average amount processed across all customer accounts over the last four weeks

10.27%

mom_growth

The percentage change in performance or volume compared to the previous month

2.27%

Wow_growth

The percentage change in performance or volume compared to the previous week

2.18%

CAC_growth

The rate of change in the CAC(How much it costs to gain a new customer

Performance Metrics

Utilization Ratio

Average utilization: 0.27

Shows how much of available credit customers are using

Customer Churn

Approximately 5,000 customers identified

Attrition analysis helps retention strategies

Delinquency rate

Delinquency_rate : 6.06%

Income Credit Correlation

Income_credit_correlation : 0.64%

What We Accomplished

Generate Dax Measures

Creates measures which shows the performance of credit card and identified customers engagement

Risk Identification

Identified risky customers using scoring model

Behavior Analysis

Analyzed customer credit behavior patterns

Decision Insights

Provided data-driven insights for strategic decisions

Limitations

Dataset Constraints

Limited to available variables in current data sources

External Factors

Economic conditions and market changes not included

Behavioral Changes

Customer behavior over time not fully captured

Historical Assumptions

Model assumes historical trends will continue

Future Opportunities

Machine Learning

Predict loan defaults using advanced algorithms

Customer Segmentation

Apply clustering techniques for better targeting

Forecasting

Predict future transaction growth patterns

Additional Future Opportunities

Fraud Detection

Implement real-time systems to identify suspicious transactions

Real-Time Integration

Connect live data feeds for immediate insights

Recommendations

01

Reduce Credit Limits

Adjust limits for high-risk customers to minimize exposure

02

Combat Churn

Design targeted offers to retain at-risk customers

03

Monitor Utilization

Track customers with high utilization ratios closely

04

Improve Satisfaction

Address lower scores among platinum card users

05

Adjust Policies

Refine credit policies based on income level correlations

Conclusion

This project demonstrates how Power BI can be applied to real banking scenarios, enabling financial institutions to analyze performance, assess customer risk, and make better credit decisions.

Data-driven insights improve profitability and reduce financial risk.